

3F7 Information Theory and Coding

Full Technical Report

Liu Zihe
zl559

University of Cambridge
December 6 2025

1 Introduction

Information theory provides the mathematical framework for quantifying information and establishing fundamental limits on data compression. Shannon’s Source Coding Theorem establishes the entropy $H(X)$ as the lower bound for the average code length of an independent and identically distributed (i.i.d.) source. While i.i.d. models are useful first approximations, real-world data sources—such as natural language—exhibit strong temporal dependencies (memory) and non-uniform distributions.

This report extends the investigation of the 3F7 short laboratory in two directions. First, we analyse **discrete stationary sources**, relaxing the i.i.d. assumption to model sources with memory. We investigate the *entropy rate* H_∞ of a stochastic process and the convergence properties of block entropies. Second, we explore **adaptive coding**, where the probability model is learned online instead of 2 separate phases (building the probability model from the complete text and encoding). We derive the Laplace estimator from Bayesian principles and evaluate the performance of adaptive arithmetic coding against static methods.

2 Overview of Short Lab

The short laboratory focused on compressing `hamlet.txt` (approx. 207kB using fixed-length ASCII encoding) using three classical techniques designed for i.i.d. sources.

Shannon-Fano Coding constructs prefix-free codes by assigning lengths $l_i = \lceil \log_2(1/p_i) \rceil$. While it guarantees unique decodability (via the Kraft inequality $\sum 2^{-l_i} \leq 1$), it is suboptimal because it ignores the combinatorial nature of packing the prefix-free code tree (codewords can still be placed at shorter leaf nodes, hence optimizing our encoding)

Huffman Coding is the optimal symbol-by-symbol prefix code. It builds a binary tree bottom-up by iteratively merging the two least probable symbols. Our experiments confirmed that its expected length L satisfies $H(X) \leq L < H(X) + 1$. However, its performance is constrained by the requirement that each symbol must use an integer number of bits.

Arithmetic Coding overcomes the integer constraint by encoding entire sequences as sub-intervals of $[0, 1)$. By mapping a sequence x^n to an interval of width $P(x^n)$, it achieves a code length $L \approx -\log_2 P(x^n)$ bits. For large n , the per-symbol length converges to the entropy $H(X)$. This method proved superior for longer blocks but required accurate probability models.

The limitation of all three methods in the short lab was the assumption that text is a memoryless source, ignoring the significant redundancy in English grammar and spelling.

3 Coding for Discrete Stationary Sources

3.1 Theoretical Background

We model natural language text as a sequence of random variables, specifically a *discrete stochastic process*, $\mathcal{X} = \{X_i\}_{i=1}^{\infty}$. We distinguish between two primary classes of sources:

Memoryless Sources (i.i.d.): The simplest model assumes that successive symbols are i.i.d. A source is *memoryless* if the joint probability of a sequence factorizes completely into the product of its components:

$$\mathbb{P}(X_1, \dots, X_n) = \prod_{i=1}^n \mathbb{P}(X_i). \quad (1)$$

For such a source, the information content is additive, and the entropy of a block is simply $H(X^n) = nH(X)$. The optimal code length per symbol is thus fixed at the marginal entropy $H(X)$. The fixed-length ASCII and standard Huffman coding (without blocking) implicitly assume this model.

Stationary Sources (Sources with Memory): Natural languages like English are not memoryless; the letter 'h' is much more likely to follow 't' than 'q'. To capture this, we relax the independence assumption while retaining time-invariance. A process is *strictly stationary* if the joint distribution of any subsequence is invariant to time shifts. That is, for any shift τ and block length n :

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \mathbb{P}(X_{1+\tau} = x_1, \dots, X_{n+\tau} = x_n). \quad (2)$$

Stationarity implies that marginal probabilities are constant $H(X_i) = H(X_1)$ and conditional probabilities depend only on the relative history, not the absolute time.

Entropy Rate: For a source with memory, the marginal entropy $H(X_1)$ overestimates the true information content because it ignores the predictability provided by the context. The fundamental limit is instead given by the *entropy rate* H_{∞} , defined as the asymptotic per-symbol entropy [1, Def 4.1]:

$$H_{\infty} = \lim_{n \rightarrow \infty} \frac{1}{n} H(X_1, \dots, X_n). \quad (3)$$

This rate represents the *irreducible randomness* of the process after all past history is accounted for. To estimate H_{∞} , we define two sequences of block entropies:

$$H_n = \frac{1}{n} H(X_1, \dots, X_n) \quad (\text{Entropy per symbol of } n\text{-block}) \quad (4)$$

$$H_n^* = H(X_n | X_{n-1}, \dots, X_1) \quad (\text{Conditional entropy}) \quad (5)$$

3.2 Convergence of Entropy Rates (Theorem 1)

For stationary sources, these sequences converge to the entropy rate H_{∞} as $n \rightarrow \infty$. We provide proofs for the properties stated in the FTR handout, referencing Cover & Thomas [1, Chap 4].

Statement 1: $H_{n+1}^* \leq H_n^*$ (Conditional entropy is non-increasing).

Proof:

$$\begin{aligned} H_{n+1}^* &= H(X_{n+1} | X_n, \dots, X_2, X_1) \\ &\leq H(X_{n+1} | X_n, \dots, X_2) \quad (\text{Conditioning reduces entropy}) \\ &= H(X_n | X_{n-1}, \dots, X_1) \quad (\text{By Stationarity}) \\ &= H_n^* \quad \square \end{aligned}$$

Statement 2: $H_{n+1} \leq H_n$ (Per-symbol entropy is non-increasing).

Proof: By the chain rule, $H(X_1, \dots, X_{n+1}) = H(X_1, \dots, X_n) + H(X_{n+1}|X_1, \dots, X_n)$. Then

$$H_{n+1} = \frac{H(X_1, \dots, X_{n+1})}{n+1} = \frac{nH_n + H_{n+1}^*}{n+1} \quad (6)$$

Since $H_{n+1}^* \leq H_n^*$ (from Statement 1), $H_n^* \leq H_n$ (from statement 3), then $H_{n+1}^* \leq H_n^* \leq H_n$ where H_n is the average of joint entropy across the n terms. Adding a term H_{n+1}^* that is smaller than or equal to the previous average must reduce (or maintain) the new average. Thus $H_{n+1} \leq H_n$. $\square$

Statement 3: $H_n^* \leq H_n$.

Proof: From the Chain Rule for entropy,

$$H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i|X_{i-1}, \dots, X_1). \quad (7)$$

Dividing by n :

$$H_n = \frac{1}{n} \sum_{i=1}^n H_i^*. \quad (8)$$

Since H_i^* is a non-increasing sequence (Statement 1), the current term H_n^* is less than or equal to all preceding terms H_i^* for $i < n$. Therefore, the average H_n must be greater than or equal to the minimum term H_n^* . $\square$

Statement 4: $\lim_{n \rightarrow \infty} H_n = \lim_{n \rightarrow \infty} H_n^* = H_\infty$.

Proof: Since H_n^* is non-increasing (Statement 1) and bounded below by 0 (entropy is non-negative), the limit $\lim_{n \rightarrow \infty} H_n^* = H'$ exists. Since H_n is the arithmetic mean (Cesàro mean) of the sequence H_n^* as $n \rightarrow \infty$, by the *Cesàro Mean Theorem* (if $a_n \rightarrow a$, then $\frac{1}{n} \sum a_i \rightarrow a$), the sequence of averages converges to the same limit [1, Thm 4.2.1]. This limit is defined as the *entropy rate* H_∞ . $\square$

3.3 Experimental Results: Entropy Rate Convergence

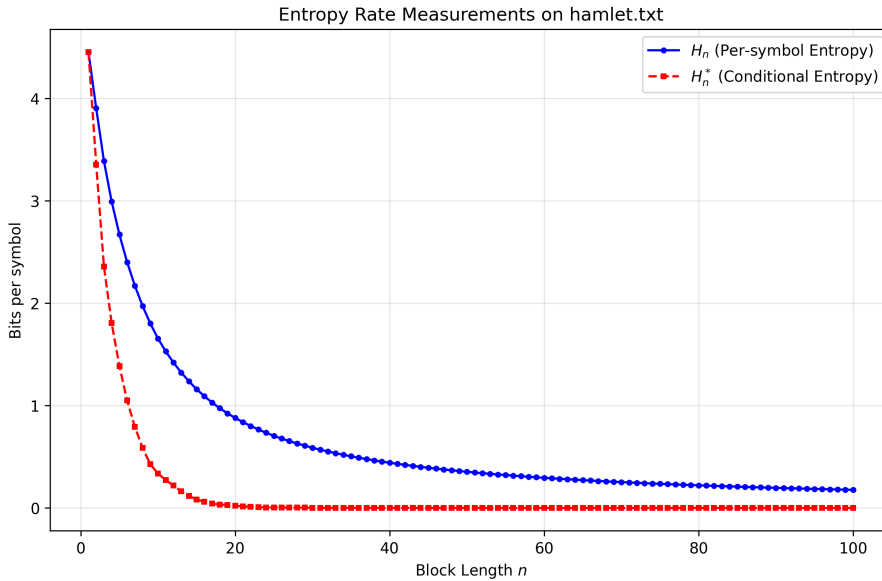


Figure 1: Entropy rate measurements on `hamlet.txt`. The estimates converge to 0 for large n due to data sparsity, not because the text is deterministic.

We measured the block entropy per symbol, $H_n = \frac{1}{n}H(X_1, \dots, X_n)$, for $n = 1$ to 100. As shown in Fig. 1, estimates for small n ($n = 1 - 20$) decrease quickly, correctly reflecting that English text is a source with memory; knowledge of previous characters reduces the uncertainty of the current one. However, for $n \geq 4$, the estimate continues to decay toward zero. The true entropy rate of English (H_∞) is surely not zero. Rather, this decay is a statistical artifact of a maximum likelihood estimator (of the entropy rate) operating in a high-dimensional regime with insufficient data (the "curse of dimensionality"). Accurate measurement of H_∞ is impossible without a sample size N that scales as $|\mathcal{X}|^n$.

3.4 Implications for Static Coding

The convergence failure observed in highlights a critical limitation for static source coding. To approach the entropy rate of a stationary source, the encoder must exploit a context of length n . In a static framework, this requires transmitting the probability model (e.g., a frequency table) to the decoder as side information. The size of this model is proportional to $|\mathcal{X}|^n$. As we increase n to capture longer dependencies, this overhead grows exponentially, quickly exceeding the size of the compressed data itself. Consequently, static methods cannot simply let $n \rightarrow \infty$; they are fundamentally limited to a small, finite n .

3.5 Context-Adaptive Arithmetic Coding

To overcome the static overhead limit, we employ *Context-Adaptive Arithmetic Coding*, often formalized as *Prediction by Partial Matching* (PPM). Instead of transmitting a massive static table, the model consists of a set of conditional probability distributions $P(X_t|C)$ that are built adaptively during encoding, where C is the context ($X_{t-n}, \dots, X_{t-1}$).

This approach directly addresses the data sparsity problem. When a symbol appears in a context C that has zero frequency (has never been seen before), encoder and decoder then "back off" to a shorter context of length $n - 1$, recursing down to order-0 if necessary. By using this escape mechanism, the algorithm can exploit very large n for frequent patterns (where statistics are rich) while falling back to shorter contexts for rare sequences. This allows the compression rate to approach H_∞ without the prohibitive $|\mathcal{X}|^n$ overhead required by static methods.

4 Adaptive Coding

4.1 Bayesian Derivation of the Laplace Estimator

In adaptive coding, we face the "zero-frequency problem": if a symbol x has not yet appeared in the stream, its Maximum Likelihood Estimate (MLE) is $P(x) = 0$. This is catastrophic for arithmetic coding. To solve this, we must assign probability mass to unseen events. We derive the solution using Bayesian inference, which treats the unknown probabilities not as fixed values, but as random variables describing our uncertainty [3, p. 97].

4.1.1 The Beta Posterior

Consider a binary version of the problem where we simply estimate the probability θ that the next symbol is a specific character 'x'. We start with **Bayes' Theorem**:

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)} \quad (9)$$

where D is the observed data (sequence of symbols), $P(\theta)$ is the *prior* belief, and $P(D|\theta)$ is the *likelihood* of the data.

- **Prior:** Since we assume no prior knowledge about the text, we must choose a prior that represents total ignorance. For a probability parameter $\theta \in [0, 1]$, the natural choice is the Uniform Distribution, also known as the $Beta(1, 1)$ distribution [2, Ch 3]. In the Bayesian framework, this corresponds to a prior which expresses no initial preference for one parameter set over another:

$$P(\theta) = 1 \quad \text{for } 0 \leq \theta \leq 1 \quad (10)$$

- **Likelihood:** Suppose we observe a stream of length n containing n_x occurrences of 'x' and n_y occurrences of other symbols ($n_y = n - n_x$). Assuming independence given θ , the probability of seeing this specific sequence is:

$$P(D|\theta) = \theta^{n_x} (1 - \theta)^{n_y}. \quad (11)$$

- **Posterior:** Multiplying the prior and likelihood gives us the posterior distribution (proportionality ignores the normalizing constant $P(D)$):

$$P(\theta|D) \propto \theta^{n_x} (1 - \theta)^{n_y} \cdot 1. \quad (12)$$

This resulting distribution is a Beta distribution with parameters $(n_x + 1, n_y + 1)$. (The general form of the Beta probability density function is: $f(\theta; \alpha, \beta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$) As n grows, this curve becomes a sharp peak around the empirical frequency n_x/n , representing our growing confidence.

4.1.2 Laplace's Rule of Succession (Binary Case)

To encode the *next* symbol X_{n+1} , we do not simply use the most probable value of θ (the peak). Instead, we must average over *all possible* values of θ , weighted by their posterior probability - a process known as *marginalization*. This predictive probability is equivalent to the expected value of the posterior distribution:

$$P(X_{n+1} = x|D) = \int_0^1 P(X_{n+1} = x|\theta) P(\theta|D) d\theta \quad (13)$$

$$= \int_0^1 \theta \cdot P(\theta|D) d\theta \quad \text{since } P(X_{n+1} = x|\theta) = \theta \text{ by definition} \quad (14)$$

$$= \mathbb{E}[\theta|D] \quad (15)$$

This expectation corresponds to the Minimum Mean-Squared Error (MMSE) estimator in Bayesian theory. Since our posterior is a Beta distribution $Beta(\alpha, \beta)$ with $\alpha = n_x + 1$ and $\beta = n_y + 1$, we use the known formula for the mean of a Beta distribution, $\mu = \frac{\alpha}{\alpha + \beta}$ [2, 4]:

$$\hat{p}(x) = \mathbb{E}[\theta|D] = \frac{\alpha}{\alpha + \beta} = \frac{n_x + 1}{(n_x + 1) + (n_y + 1)} = \frac{n_x + 1}{n + 2}. \quad (16)$$

This result is Laplace's Rule of Succession. It effectively adds 1 "pseudocount" to both outcomes, preventing zero probabilities [2, p. 52].

4.1.3 Generalization to Text (The Dirichlet Distribution)

We now generalize from binary outcomes to an alphabet of size K (e.g., $K = 256$ for extended ASCII). The unknown parameter is now a probability vector $\theta = (\theta_1, \dots, \theta_K)$ where $\sum_{i=1}^K \theta_i = 1$.

The natural generalization of the Beta distribution to K dimensions is the *Dirichlet distribution* [2, pg 317]. Just as the Beta distribution represents our uncertainty about a single probability

parameter, the Dirichlet distribution represents our uncertainty about a *vector* of probabilities that must sum to 1. The Dirichlet distribution with parameters $\alpha = (\alpha_1, \dots, \alpha_K)$ has probability density [2, Eq. 23.30]:

$$P(\theta|\alpha) = \frac{1}{B(\alpha)} \prod_{i=1}^K \theta_i^{\alpha_i-1} \quad (17)$$

where $B(\alpha) = \frac{\prod_i \Gamma(\alpha_i)}{\Gamma(\sum_i \alpha_i)}$ is the multivariate Beta function (normalizing constant) and each $\alpha_i > 0$ [5].

Suppose we observe a sequence with symbol counts $\mathbf{n} = (n_1, \dots, n_K)$ where $N = \sum_i n_i$ is the total number of observations.

- **Prior:** We start with a uniform prior representing complete ignorance, which corresponds to $\alpha = (1, \dots, 1)$:

$$P(\theta) = \text{Dir}(1, \dots, 1) = \text{Uniform} \quad (18)$$

- **Likelihood:** The probability of observing these counts follows the multinomial distribution:

$$P(\mathbf{n}|\theta) \propto \prod_{i=1}^K \theta_i^{n_i} \quad (19)$$

- **Posterior:** By Bayes' theorem, the posterior is also Dirichlet with updated parameters:

$$P(\theta|\mathbf{n}) = \text{Dir}(\alpha_1 + n_1, \dots, \alpha_K + n_K) = \text{Dir}(1 + n_1, \dots, 1 + n_K) \quad (20)$$

To predict the next symbol, we must marginalize over all possible parameter values weighted by the posterior. This predictive probability equals the *mean* of the posterior Dirichlet distribution. For a Dirichlet distribution with parameters $(\alpha'_1, \dots, \alpha'_K)$, the mean is [2, Eq. 23.32]:

$$\mathbb{E}[\theta_i] = \frac{\alpha'_i}{\sum_{j=1}^K \alpha'_j} \quad (21)$$

Substituting our posterior parameters $\alpha'_i = 1 + n_i$:

$$\hat{p}(i) = \mathbb{E}[\theta_i|\mathbf{n}] = \frac{n_i + 1}{\sum_{j=1}^K (n_j + 1)} = \frac{n_i + 1}{N + K} \quad (22)$$

This is the *Laplace estimator*. It ensures every symbol has non-zero probability of at least $1/(N + K)$, solving the zero-frequency problem. The uniform prior $\alpha = (1, \dots, 1)$ effectively adds 1 "pseudocount" to each symbol, which can be interpreted as pretending we observed each symbol once before seeing any real data [5, 2].

4.2 Experimental Performance

We estimate the performance of the adaptive arithmetic code.

Convergence: Fig. 2 shows the running average code length L/n converging to the static entropy H_1 . The "learning penalty" (excess code length) decreases as $O(1/n)$, consistent with universal coding theory [1, Sec 13.2].

Bias Parameter δ : We varied δ (Fig. 3) and found the optimal range $\delta \in [0.5, 1.0]$.

- Small δ (< 0.1): Overfitting. Unseen symbols have near-zero probability, causing massive penalties.

- Large δ (> 5): Underfitting. The strong prior washes out the observed data, slowing adaptation.

Adaptive vs. Non-Adaptive: Fig. 4 compares total bits. Non-adaptive coding requires overhead to store the frequency table ($\approx K \log n$ bits). Adaptive coding has no header but higher initial redundancy. The crossover point for Hamlet is at $n \approx 50,000$ symbols. For files shorter than this, adaptive coding is more efficient.

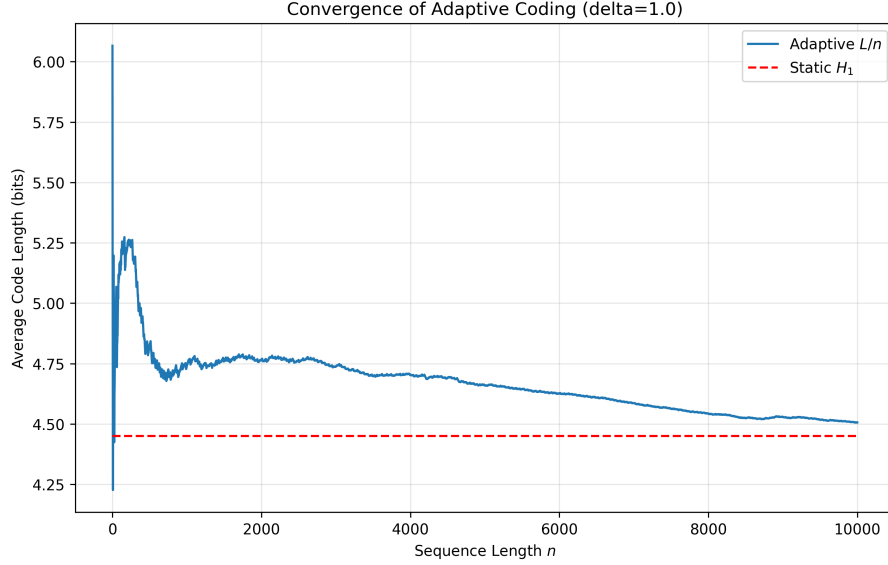


Figure 2: Convergence of adaptive coding rate (L/n) to the empirical entropy (H_1) for $\delta = 1$. The overhead vanishes as the model “learns” the source statistics.

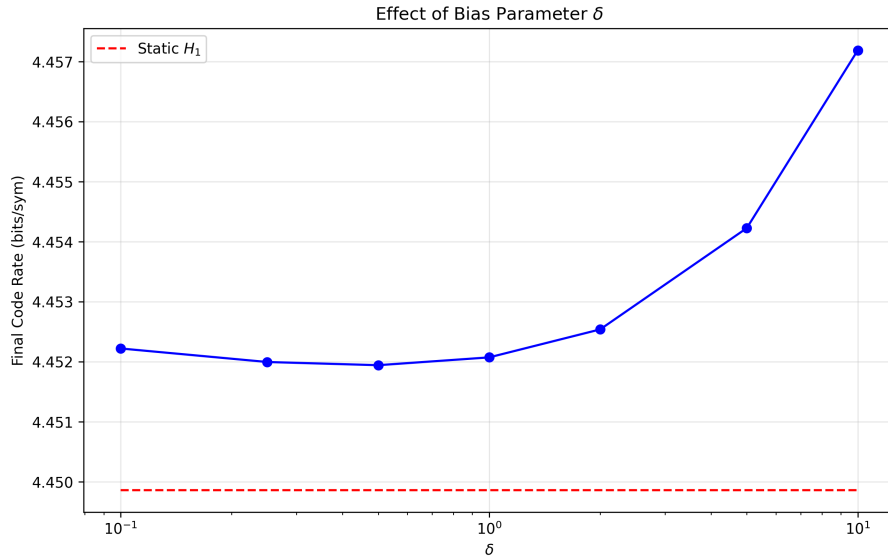


Figure 3: The effect of the bias parameter δ . The minimum code length is achieved around $\delta \approx 1.0$, corresponding to the Laplace estimator.

5 Conclusion

This report demonstrated that while English text is a stationary source with memory, exploiting this theoretically (via H_∞) is hindered by sparse data. Adaptive coding offers a robust solution,

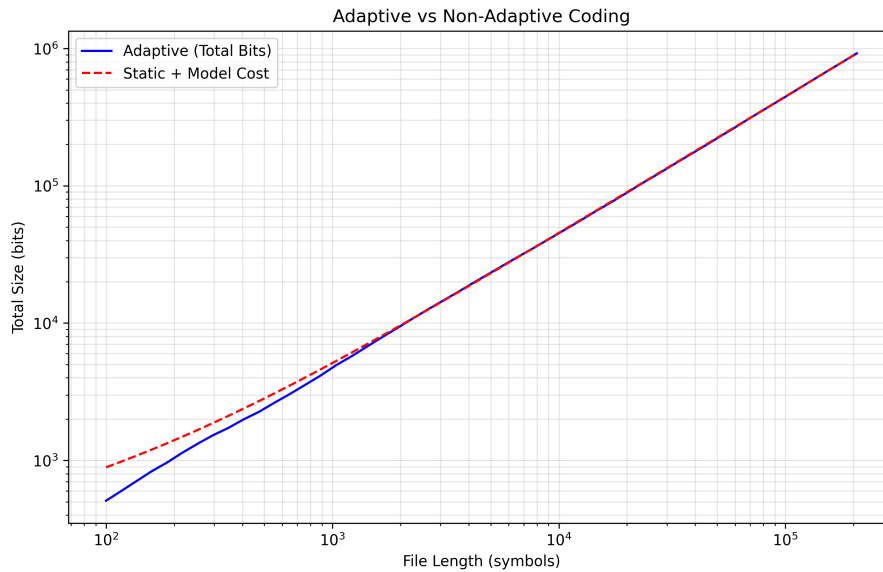


Figure 4: Total bits required for Adaptive vs Non-Adaptive coding. The crossover at $n \approx 50,000$ highlights the efficiency of adaptive coding for short streams.

allowing compression to approach the source entropy without prior knowledge of statistics. By deriving the Laplace estimator from Bayesian principles, we showed that adding a “pseudo-count” of 1 is not heuristic but the optimal strategy under a uniform prior assumption. This principle underpins modern compressors like LZ77 and prediction models in Large Language Models.

References

- [1] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley-Interscience, 2006.
- [2] D. J. C. MacKay, *Information Theory, Inference, and Learning Algorithms*. Cambridge University Press, 2003.
- [3] S. Godsill, *Estimation Theory and Inference Methods*, Lecture Notes (3F3), University of Cambridge, 2023
- [4] J. Weisberg, “Laplace’s Rule of Succession,” *Formal Epistemology*, 2019. Available: <https://jonathanweisberg.org/post/inductive-logic-2/>
- [5] A. Tsun, “The Beta and Dirichlet Distributions,” Chapter 7.4 in *Probability & Statistics with Applications to Computing*, CS109 Course Reader, Stanford University, 2025. Available: https://web.stanford.edu/class/archive/cs/cs109/cs109.1218/files/student_drive/7.4.pdf
- [6] E. P. Xing, “Bayesian Nonparametrics: Dirichlet Processes,” Lecture 19 in *10-708: Probabilistic Graphical Models*, Lecture notes by C. Malings and J. Gao, Carnegie Mellon University, Spring 2014. Available: https://www.cs.cmu.edu/~epxing/Class/10708-14/scribe_notes/scribe_note_lecture19.pdf